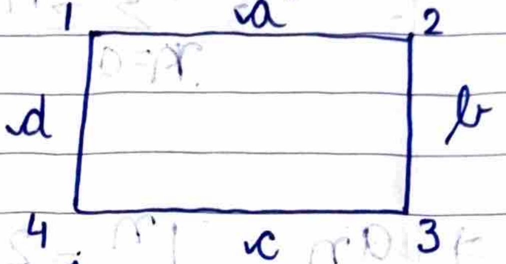
Unit → 3

→ **Graphs**

Graph is a set of (V, E) where V is a set of vertices & E is a set of edges.

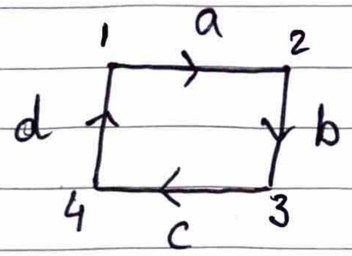
V = {1, 2, 3, 4}
E = {a, b, c, d}



→ **Undirected Graph**

No directions of edge is ~~spcified~~ specified

→ **Directed Graph**



The directions of edges are specified

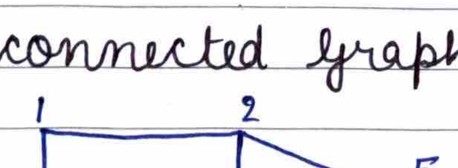
→ **Degree of vertex**

No of edges incident with a vertex

degree (1) = 2

For directed graphs:
- Indegree (1) = 1 (comes towards 1)
- Outdegree (1) = 1 (away from 1)

→ **Disconnected Graph**



When we can't go from one vertex to other

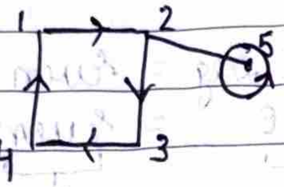
Pendant vertex

↳ degree (P.V) = 1

counted in both in & out degrees

→ Loop (Degree of a loop = 2)

An edge starting and ending at same pt. (vertex)



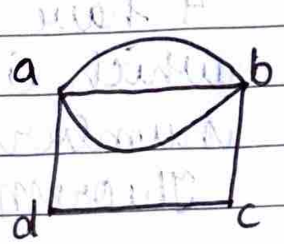
deg (5) = 3
(2 for loop & 1 for □)

→ Simple graph

The graph in which there's single edge b/t any two vertices

→ Multi graph

More than one edge between two vertices



3 edges b/t a & b

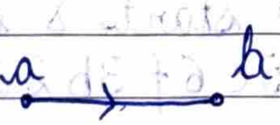
→ Theorem

~~Total no of edges in a graph is always even~~

The sum of degree of all vertices in a graph is even.

• Proof :-

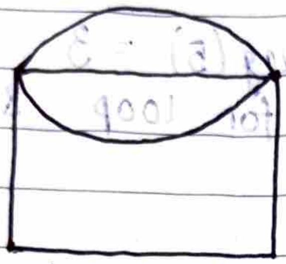
Whenever an edge is incident to a vertex it's degree is counted twice once while leaving & other while coming so it's always twice the no of edges. so it's even.



for a → (out)

for b → (in)

→ Prove that no of vertices of odd degree in a graph is always even



Total deg = Even deg + Odd deg

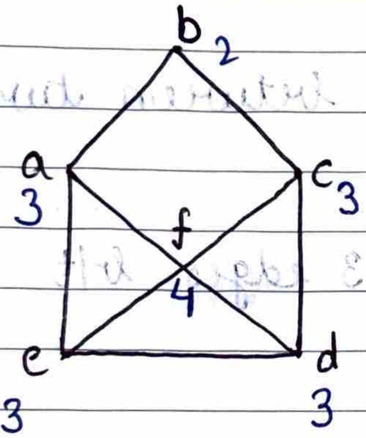
$\underbrace{2 \times e}_{\text{even}} = \underbrace{\text{Even}}_{\text{even}} + \text{odd deg}$

e → total edges

Even = Even + Even

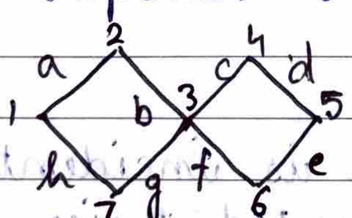
Hence, no of vertices having odd degree must be an even number.

→



4 have odd degree which is an even number
Theorem satisfies

→ Walk in a graph
It's a sequence in a graph of vertices and edges to move from one vertex to another using repeated vertices & edges.



1, a, 2, b, 3, c, 4, d, 5, e, 6, f, 3, b, 2

Open walk → vertex 1 → vertex 2

Closed Walk → starts & ends at same vertex

1 a 2 b 3 c 4 d 5 e 6 f 3 b 2 a 1

→ **Trail**

It can have a repeated vertex but not a repeated edge

eg → 1a 2b 3c 4d 5e 6f 3g 7

→ **Path**

Path shouldn't consist of any of repeated vertex or repeated edge

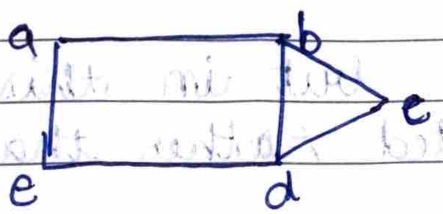
1a 2b 3c, 4d 5e

→ **Cycle**

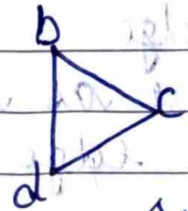
A closed path is called cycle

→ **Subgraph**

$G' = (V', E')$ is said to be a subgraph of G if $V' \subseteq V$ & $E' \subseteq E$ incident only with vertices in V'



Graph

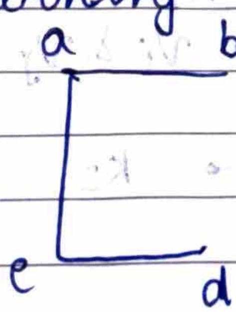


Subgraph

$V = \{a, b, c, d, e\}$

$V' = \{b, c, d\}$

→ Compliment of a subgraph is remaining subgraph which completes the graph on combining with other subgraph



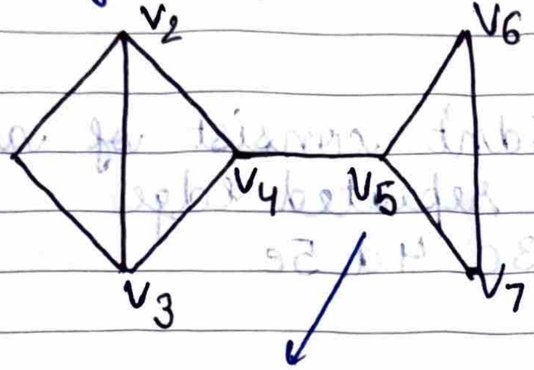
→ Spanning subgraph

A sub graph which consists all vertices of the graph but not edges

→ Distance & Diameter

For any vertices v_1 & v_2

- The shortest path b/t them is distance and longest path is diameter



→ Cutpoints   v5 is a cut point

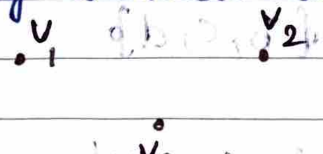
If deleting a vertex leaves the graph disconnected then that vertex is a cutpoint. While deleting any vertex also remove all edges joint to it

→ Bridge

Same as cutpoints, but in this case an edge is deleted rather than vertex

→ Null graph

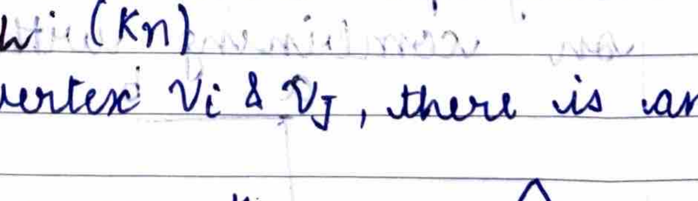
It has only vertices and no edges (each vertex is isolated i.e. degree is zero)



→ Complete graph (K_n)

If for every vertex v_i & v_j, there is an edge in graph.

If $n = 2$,



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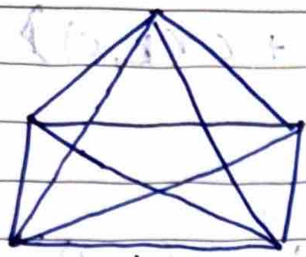
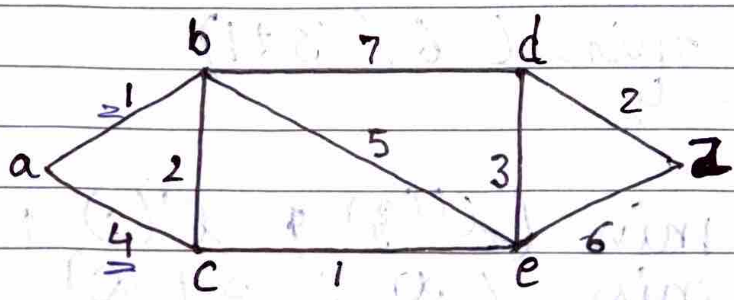


In complete graph K_n every vertex has degree $= n-1$

→ Weighted graph
If we assign weights to edges

→ Dijkstra's algorithm for shortest distance b/t 2 vertices in weighted graph (a)

1. Initially $P = \{a\}$, $T = \{V\} - \{a\}$
   ↳ set of all vertices
2. For vertex t in T, let $l(t) = w(a, t)$
3. Select ~~vertices~~ vertex that has smallest index. Let x is the vertex
4. If x is that vertex move it from set T to set P & stop
   If not, $P' = P \cup \{x\}$, $T' = T - \{x\}$
   For every vertex t in T', complete the index
   $$l'(t) = \min[l(t), l(x) + \omega(x,t)]$$
5. Repeat steps 3 & 4



shortest dist b/t a & z

$P = \{a\}$ $\quad T = \{b, c, d, e, z\}$

$l(b) = 1 \quad l(c) = 4 \quad l(d) = l(e) = l(z) = \infty$

$P' = \{a, b\} \quad T' = \{c, d, e, z\}$

$$l'(c) = \min(l(c),\ l(b) + w(b,c))$$
$$= \min(4,\ 1+2)$$
$$= 3$$

$$l'(d) = \min(l(d),\ l(b) + w(b,d))$$
$$= \min(\infty,\ 1+7)$$
$$= 8$$

$$l'(e) = \min(l(e),\ l(b) + w(b,e))$$
$$= \min(\infty,\ 1+5)$$
$$= 6$$

$$l'(z) = \min(l(z),\ l(b) + w(b,z))$$
$$= \min(\infty,\ 1+\infty)$$
$$= \infty$$

$\Rightarrow P'' = \{a, b, c\}$ $\qquad T'' = \{d, e, z\}$

$$l''(d) = \min(l'(d),\ l'(c) + wt(c,d))$$
$$= \min(8,\ 3+\infty)$$
$$= 8$$

$$l''(e) = \min(l'(e),\ l'(c) + wt(c,e))$$
$$= \min(6,\ 3+1)$$
$$= 4$$

$$l''(z) = \min(l'(z),\ l'(c) + wt(c,z))$$
$$= \min(\infty,\ 3+\infty)$$
$$= \infty$$

$\Rightarrow \quad P''' = \{a, b, c, e\} \qquad T''' = \{d, z\}$

$$\ell'''(d) = \min(\ell''(d),\ \ell''(e) + wt(d, e))$$
$$= \min(8,\ 4 + 3)$$
$$= 7$$

$$\ell'''(z) = \min(\infty,\ 4 + 6)$$
$$= 10$$

$\Rightarrow \quad P'''' = \{a, b, c, e, d\} \qquad T'''' = \{z\}$

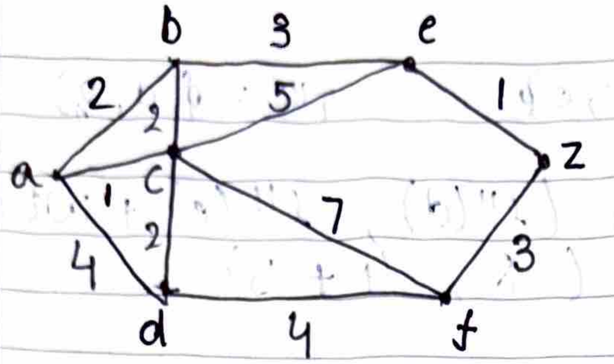
Hence shortest distance

~~a → b → c → e → d → z~~

$$\ell''''(z) = \min(\ell'''(z),\ \ell'''(d) + wt(d, z))$$
$$= \min(10,\ 7 + 2)$$
$$= 9$$

$\Rightarrow$ min dist = 9 units

Path $a \to b \to c \to e \to d \to z$



a & z

$P = \{a\}$ $\quad T = \{b, c, d, e, f, z\}$

$l(b) = 2 \quad l(c) = 1 \quad l(d) = 4$

$l(e) = \infty \quad l(f) = \infty \quad l(z) = \infty$

$P' = \{a, c\} \quad T' = \{b, d, e, f, z\}$

$l'(b) = \min(2, 1+2)$
$= 2$

$l'(d) = \min(4, 1+2)$
$= 3$

$l'(e) = \min(\infty, 1+5)$
$= 6$

$l'(f) = \min(\infty, 1+7)$
$= 8$

$l'(z) = \min(\infty, 1+\infty)$
$= \infty$

$P'' = \{a, c, b\} \quad T'' = \{d\,e\,f\,z\}$

b→d

$l''(d) = \min(3, 2+\infty)$
$= 3$

$l''(e) = \min(6, 2+3)$
$= 5$

$l''(f) = \min(8, 2+\infty)$
$= 8$

$l''(z) = \min(\infty, 2+\infty)$
$= \infty$

$P''' = \{a, c, b, d\}$ T''' $[e, f, z)$

$l'''(e) = \min\{5, 3 + \infty) = 5$

$l'''(f) = \min(8, 3+4)$
$= \cancel{6}\ 7$

$l'''(z) = \min(\infty, 3 + \infty)$
$= \infty$

$P'''' = \{a, c, b, d, e\}$ $T'''' = \{f, z\}$

$l''''(f) = \min(7, 5 + \infty)$
$= 7$

$l''''(z) = \min(\infty, 5+1)$
$= 6$

~~Path →~~ min dist. = 6

~~a → c → b → d → e → z~~

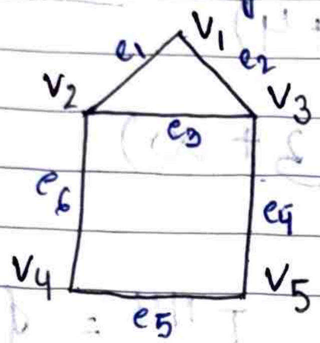
Shortest Path
a → c → b → d → e → z

→ Representation of graphs

1) Adjacency Matrix

G n vertices , n x n matrix

$A_{ij} = 1$ if v_i adjacent to v_j



$$\begin{array}{c|ccccc} & v_1 & v_2 & v_3 & v_4 & v_5 \\ \hline v_1 & 0 & 1 & 1 & 0 & 0 \\ v_2 & 1 & 0 & 1 & 1 & 0 \\ v_3 & & & & & \\ v_4 & & & & & \\ v_5 & & & & & \end{array}$$

2) Incidence matrix

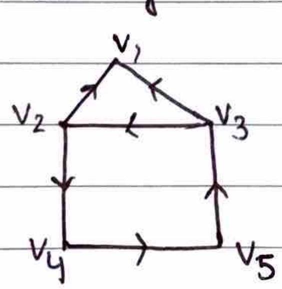
Graph G with n vertices & m edges, It has a n x m matrix

$A_{ij} = 1$ if v_i is incident with e_j

$$\begin{array}{c|cccccc} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \hline v_1 & 1 & 1 & 0 & 0 & 0 & 0 \\ v_2 & 1 & 0 & 1 & 0 & 0 & 1 \\ v_3 & & & & & & \\ v_4 & & & & & & \\ v_5 & & & & & & \end{array}$$

→ For directed graphs (or digraphs)

• Adjacency matrix



n vertices n x n matrix

$A_{ij} = 1$ if (v_i, v_j) is edge that exists

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ & & & & \\ & & & & \end{bmatrix}_{5 \times 5}$$

- Incidence matrix

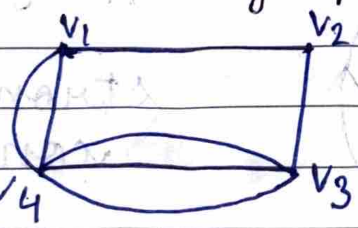
$A_{ij} = 1$ if v_i incident with e_j & is initial vertex

$A_{ij} = -1$ if v_i incident with e_j & is final vertex

$A_{ij} = 0$ otherwise

$$\begin{array}{c|cccccc} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \hline v_1 & -1 & -1 & 0 & 0 & 0 & 0 \\ v_2 & 1 & 0 & -1 & 0 & 0 & 1 \\ v_3 & 0 & 1 & 1 & 0 & -1 & 0 \\ v_4 & 0 & 0 & 0 & 0 & 1 & -1 \\ v_5 & 0 & 0 & 1 & -1 & 0 & 0 \end{array}$$

→ For multigraph → more than one edge between 2 vertices



- Adjacency graph

$A_{ij} = n$ where n is no of edges between v_i & v_j

$$\begin{array}{c|cccc} & v_1 & v_2 & v_3 & v_4 \\ \hline v_1 & 0 & 1 & 0 & 2 \\ v_2 & 1 & 0 & 1 & 0 \\ v_3 & 0 & 1 & 0 & 3 \\ v_4 & 2 & 0 & 3 & 0 \end{array}$$

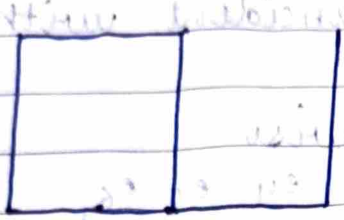
→ Weakly connected digraph

It becomes a connected graph if we ignore the directions.

→ unilaterally connected digraph

∃ a path (directed) from v_i to v_j or ~~vice versa~~ viceversa

$v_i \to v_j$



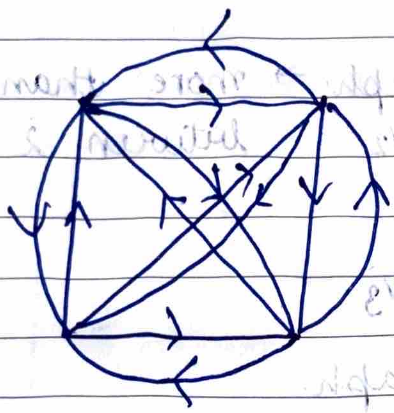
not unilaterally connected

→ strongly connected digraph

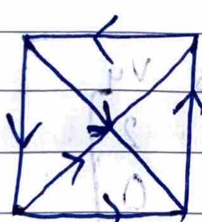
∃ a directed path from v_i to v_j and vice versa

$v_i \to v_j$

& $v_j \to v_i$



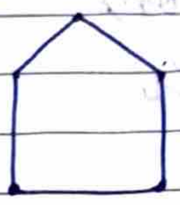
→ strongly connected



→ unilaterally connected

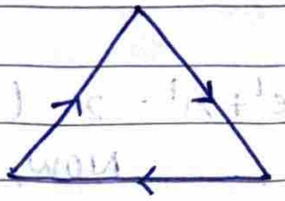
degree each vertex is same

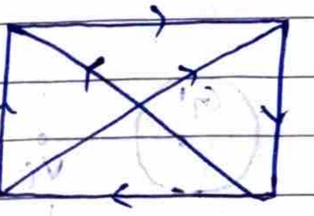
Q Draw a ② regular graph with 5 vertices
↳ degree of all vertices is = 2



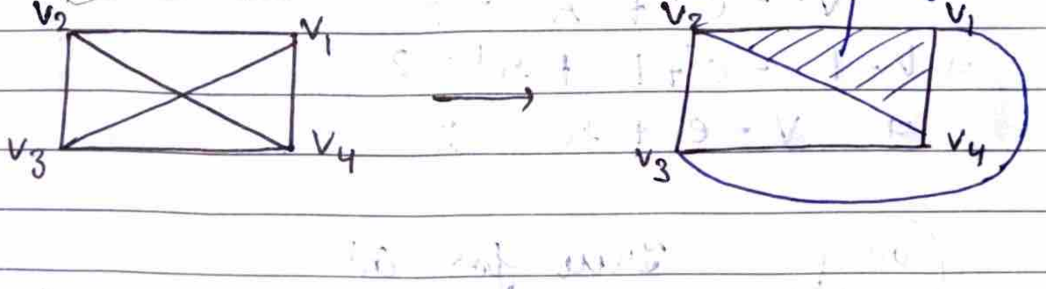
→ Directed complete graph
every vertex has an edge to every other vertex

For ~~either~~ every two vertices v_i & v_J either (v_i, v_J) or (v_J, v_i) is an edge



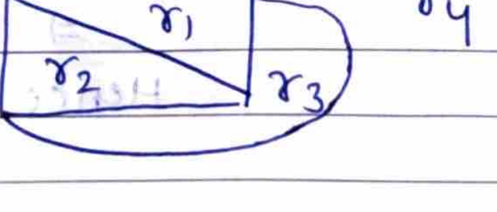


→ Planar graph
If no two edges intersect each other except at common vertices



→ Region
Area bounded by 3 or more edges

- Finite → r_1, r_2, r_3
- Infinite → r_4

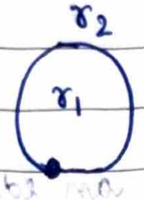


→ Euler's theorem for planar graphs

For any planar graph with v vertices, e edges and r regions.

$$v - e + r = 2$$

Proof For 1 vertex



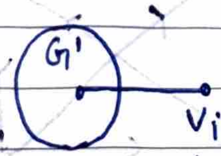
$V = 1, \ e = 1, \ r = 2$

$V + r - e = 2$ is true

For $V = 2$

$e = 1 \ r = 1$

$V - e + r = 2$ is true



$V' - e' + r' = 2$ (Hypothesis)

Now,

$1 + V' = V$

$e = e' + 1$

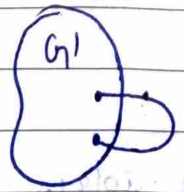
$r = r'$

This V_i wasn't included in G' initially but now we include it

$V' - e' + r' = 2$

$\Rightarrow V - 1 - e + 1 + r' = 2$

$\Rightarrow V - e + r = 2$



True for G'

$V' - e' + r' = 2$

$V = V' \quad e = e' + 1 \quad r = r' + 1$

$V - e + 1 + r - 1 = 2$

$\Rightarrow V - e + r = 2$

Hence proved

→ Properties of planar graphs

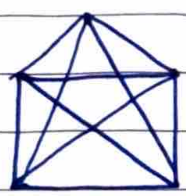
1. In a connected planar graph,

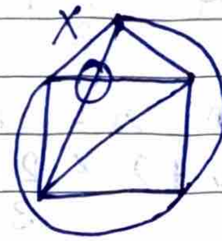
$$r \le \frac{2}{3} e$$

2. $v - e + r = 2$

3. $3V - e \ge 6$

4. A complete graph K_n will be planar if $n < 5$





its not planar whatever we do

Q P.T. complete graph K_4 is planar

K_4 has 6 edges

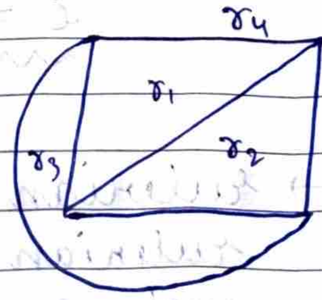
Also $3V - e \ge 6$

$3V - 6 \ge 6$

$3V \ge 12$

$V \ge 4$

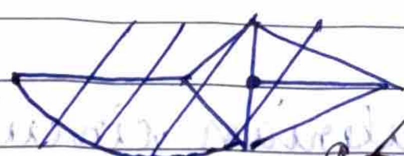
$4 \ge 4$

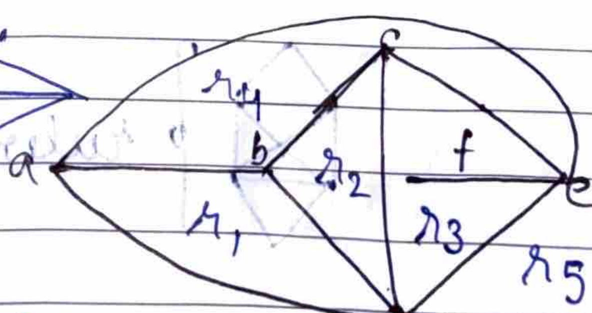


It holds true so K_4 is planar

→ Degree of the region

It's the length of closed walk of borders of a region.





$\deg(r_1) = 3$ $\quad \deg(r_3) = 5$

$\deg(r_2) = 3$ $\quad \deg(r_4) = 4$ $\quad \deg(r_5) = 3$

For n regions,

$$\sum_{i=1}^{n} \deg(r_i) = 2 \times e$$

→ PT every planar graph with v vertices and e edges and $v \geq 3$ satisfies the inequality $e \leq 3v - 6$

since $2e \geq 3r$

$$r \leq \frac{2}{3} e$$

also, $r = e - v + 2$

$$\Rightarrow e - v + 2 \leq \frac{2}{3} e$$

$$\Rightarrow 3e - 3v + 6 \leq 2e$$

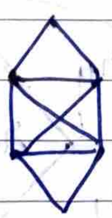
$$e \leq 3v - 6 \quad (v \geq 3)$$

→ Eulerian Path and circuit

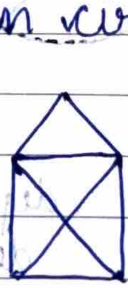
Eulerian path is a path constructed in such as way that all edges are traversed exactly one.

If eulerian path is closed path then it's a circuit.

- All vertices have even degree in eulerian circuit



} eulerian circuit



} It have eulerian path

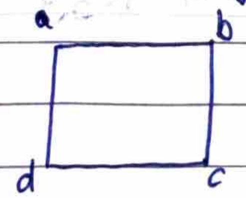
PT

→ A graph possesses a eulerian path iff it is connected & has only 2 vertices of odd degree & degree of all other vertices is even.
Let's assume the graph has eulerian path
Now degree of each vertex will be even ∵ each ~~vertex~~ edge would be counted twice while entering or leaving.

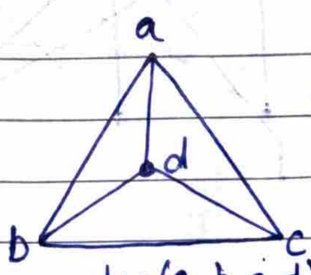
case I Assume graph has eulerian path
each edge leaves and enters vertex & contribute to their degree. so each vertex has even degree

case II Assume all vertices have even degree
Strat traversing edges of graph without repetion
If all the edges transverse so eulerian path exist otherwise it forms sub graphs of remaining points. Since graph is connected, subgraph should be connected with main point

→ Isomorphic graphs
Two graphs are isomorphic to each other if there is one to one correspondence between vertices & edges.



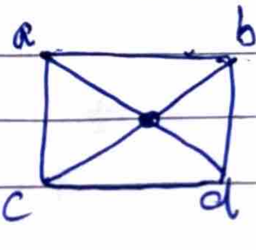
deg(a,b,c,d) = 2



deg(a,b,c,d) = 3

NOT ISOMORPHIC

Isomorphic



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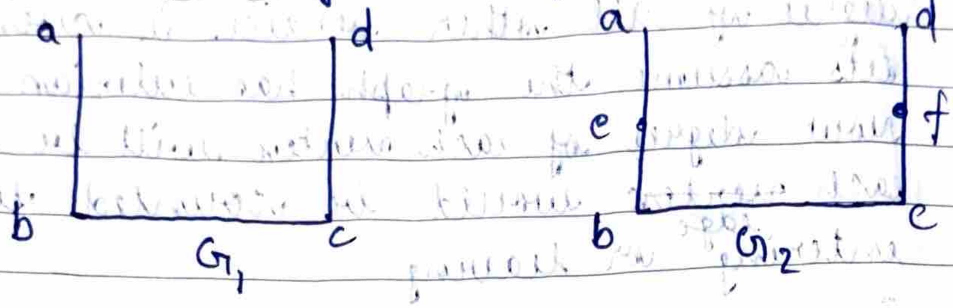
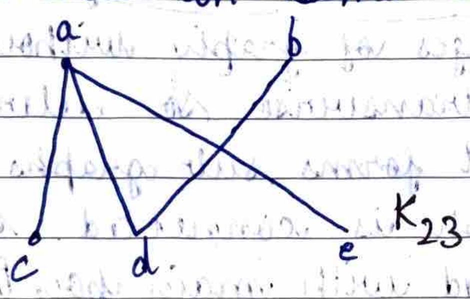
→ Homeomorphic
Subdivisions of an edge



G_1 & G_2 are homeomorphic

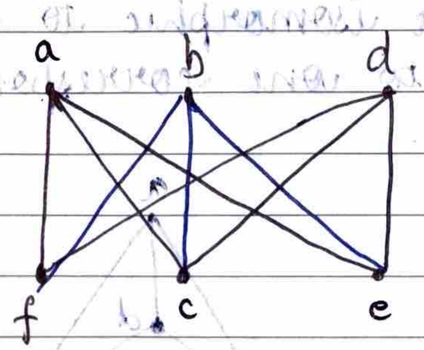
→ Bipartite graph
If we divide vertices in two disjoint subsets such that there is no edge between any 2 vertices in same set



$V_1 = \{a, b\}$
$V_2 = \{c, d, e\}$

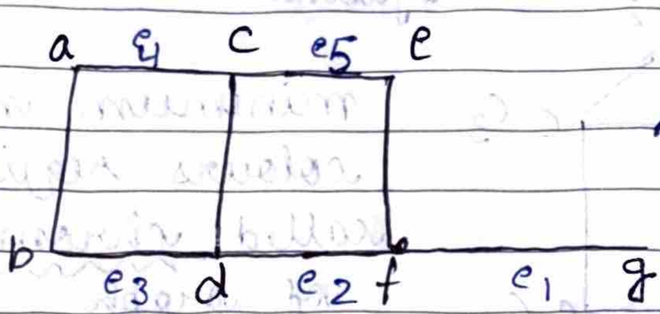
K_{23}

→ complete bipartite graph (K_{mn})



→ K_{33}

→ Hamiltonian path and circuit
A path which traverses each vertex of the graph only once



↗ Path ✓
circuit ✗

→ Theorem
∃ a hamiltonian path in a directed complete graph

Proof Let there is a path $v_1, v_2, v_3 \ldots v_i$ and it traverses all these only once. If ∃ some vertex v_x which isn't included in this path.

As this is a directed complete graph then either (v_1, v_x) is an edge or (v_x, v_1) is an edge.

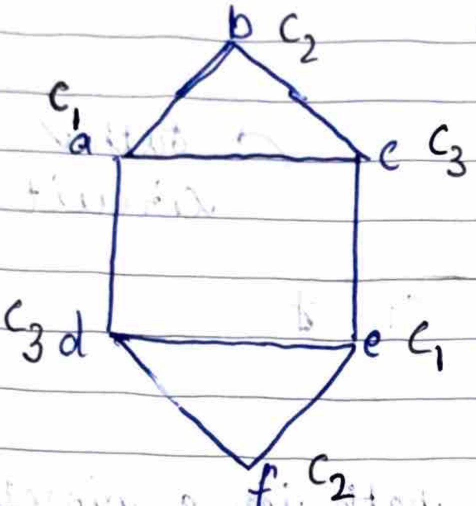
Case I $(v_1\ v_x)$ is an edge
$v_x, v_1, v_2 \ldots v_i$ → New path

Case II (v_x, v_1) is an edge
Then either $(v_2\ v_x)$ or $(v_x\ v_2)$ is an edge
$v_1\ v_x\ v_2 \ldots v_i$
Take v_3

Hence it will always become a hamiltonian path by repeating same arguement for all vertices till last vertex.

_/_/_

→ Graph coloring
assigning colors to vertices of a graph such that no two adjacent vertices have same color

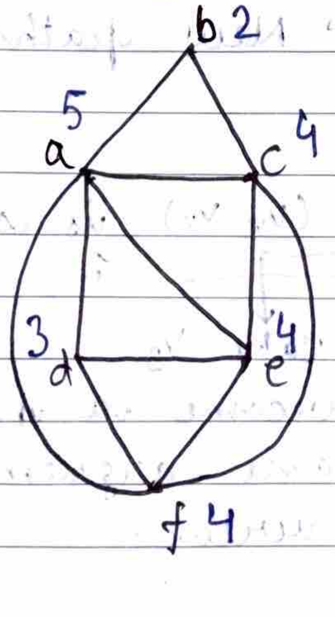


minimum no of these colours required is called chromatic no of graph $\chi(G)$

$\chi(G) = 3$

→ Welch - Powell Algorithm (Just to find no of colors [May or Mayn't give $\chi(G)$])

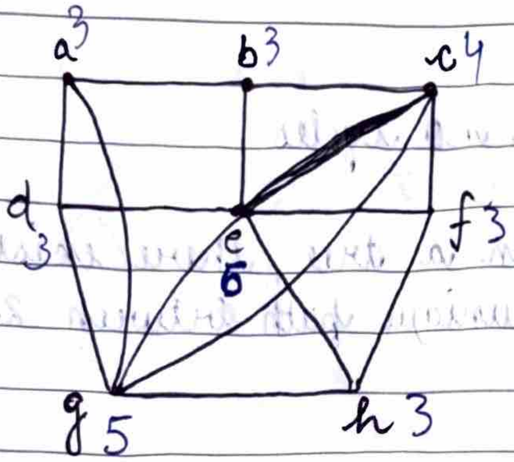
1. Acc to this we arrange all vertices in decreasing order of their degrees
2. Assign color C_1 to first vertex in this list and to subsequent vertices in list if they aint adjacent
3. Repeat same with all vertices until all vertices have been colours



Descending order
a c e f d b
C_1 C_2 C_3 C_4 C_2 C_3

4 colours are required

_/_/_



~~g e c a b h f d~~
~~c_1 c_2 c_3 c_4 c_4 c_4 c_3~~

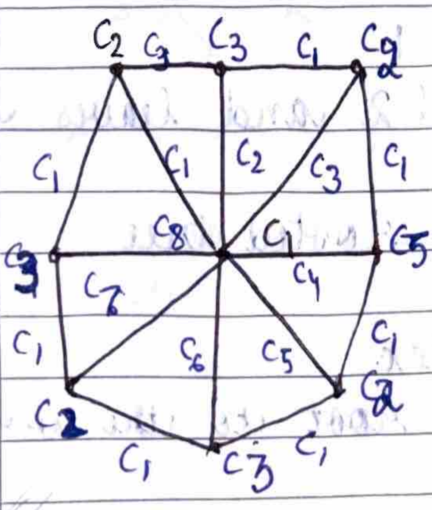
e g c a b h f d
c_1 c_2 c_2 c_1 c_3 c_3 c_2 c_3

3 colours required

→ Edge colouring

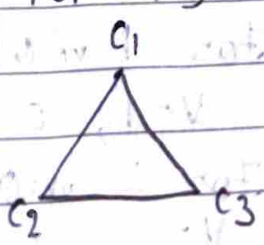
- To edges having (sharing) same vertex don't have same colour
- Min no of colours req for edge colouring is chromatic index → $\chi'(G)$

$\chi'(G)$ may/mayn't be same as $\chi(G)$

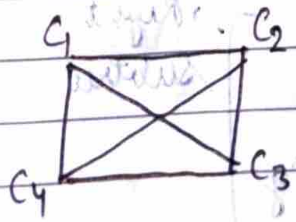


$\chi(G) = 3$
$\chi'(G) = 8$

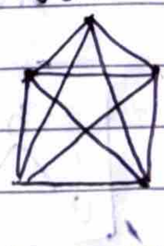
→ For K_3



For K_4



For K_5



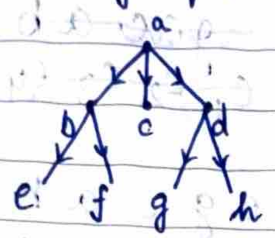
$\chi_G = 5$

For any complete graph (K_n) $\chi(G) = n$

_/_/_

→ Tree

It's a graph with no cycles



In a tree, there exists a unique path between 2 vertices

Proof: Let V_1 & V_2 have 2 paths, path-1 to go from V_1 to V_2 and path-2 to go from V_2 to V_1

That forms a closed path

⇒ Our assumption is wrong

→ Rooted tree

Exactly one vertex whose indegree is 0 and indegree of all other vertices is 1

root (a) is at level 0

b, c, d are at level 1

e, f, g, h are at level 2 and leaves of tree

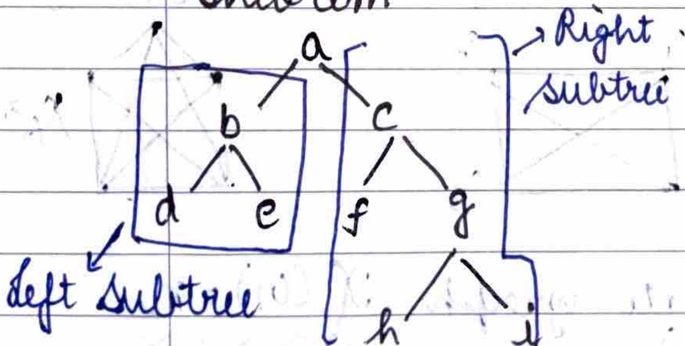
An empty tree is a rooted tree

→ Path length of a vertex

It's no of edges from root to the vertex

Path length (f) = 2

→ Theorem



$e = v - 1$

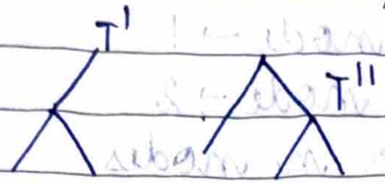
For 1 node

$v = 1, \ e = 0$

For 2 nodes

$V = 2, \ e = 1$

Divide T into T', T'' by deleting an edge



Assume theorem is true for T' & T''

$e' = v' - 1$ & $e'' = v'' - 1$

Also, $e = e' + e'' + 1$ & $v'' + v' = v$

Also, $e' + e'' = v' + v'' - 2$

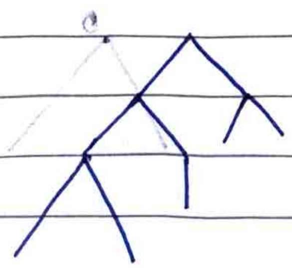
$\Rightarrow e = v - 1$

→ Binary Tree

Every vertex has 2 or less than 2 children

→ Complete Binary Tree

All the leaf nodes are on the left side of last level



→ Full Binary Tree

All the leaf nodes are at same level

→ Depth of Tree

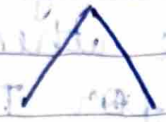
It's the maximum no of nodes in a branch of tree.

→ In a full binary tree, the no. of leaf nodes is equal to the no of internal nodes

→ In a non empty full binary tree the no. of leaf nodes is 1 more than internal nodes

_/_/_

Proof: consider 2 external nodes



Internal node → 1
External node → 2

Assume it's true for n nodes
⇒ n → internal & n+1 → external nodes

For n+1 internal nodes
external nodes will be → n + 1 + 2 − 1

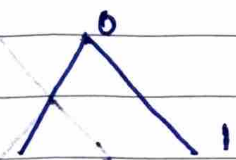
= n + 2

For 2 leaves of internal node

subtract 1 leaf node which is now internal node

→ Prove that maximum no of nodes on level n is 2^n where $n \geq 0$

For $n = 0$, nodes = 1
$2^0 = 1$ also



For $n = 1$, nodes = 2
$2^1 = 2$ also

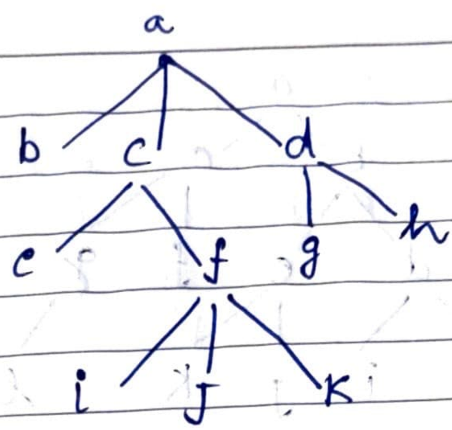
Let n−1 level has 2^{n-1} nodes.
since, it's binary tree, so each node at level n−1 will have maximum of 2 children

no. of nodes $= 2 \times 2^{n-1}$
$= 2^n$

at nth level

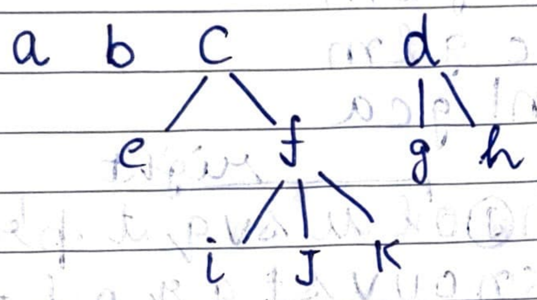
→ Tree transversals

- Preorder
- Inorder
- Postorder



→ Preorder

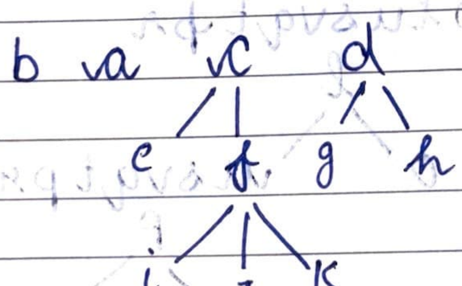
- Root → Left subtree → Right subtree

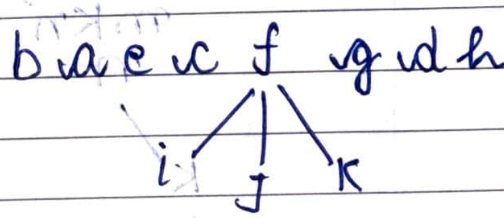


ab c efiJK d gh = abcefijkdgh

→ Inorder

Left subtree → root → Right subtree



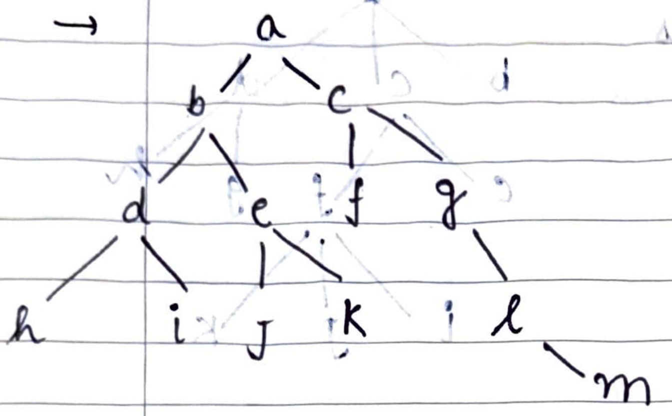


= b a e c i f J K g d h

→ Post order

Left subtree → Right Subtree → Root

b e i J K f c g h d a



Pre abdhiecfJKglm

In hdibeaJfKcglm

Post hidebJKfmlgca.

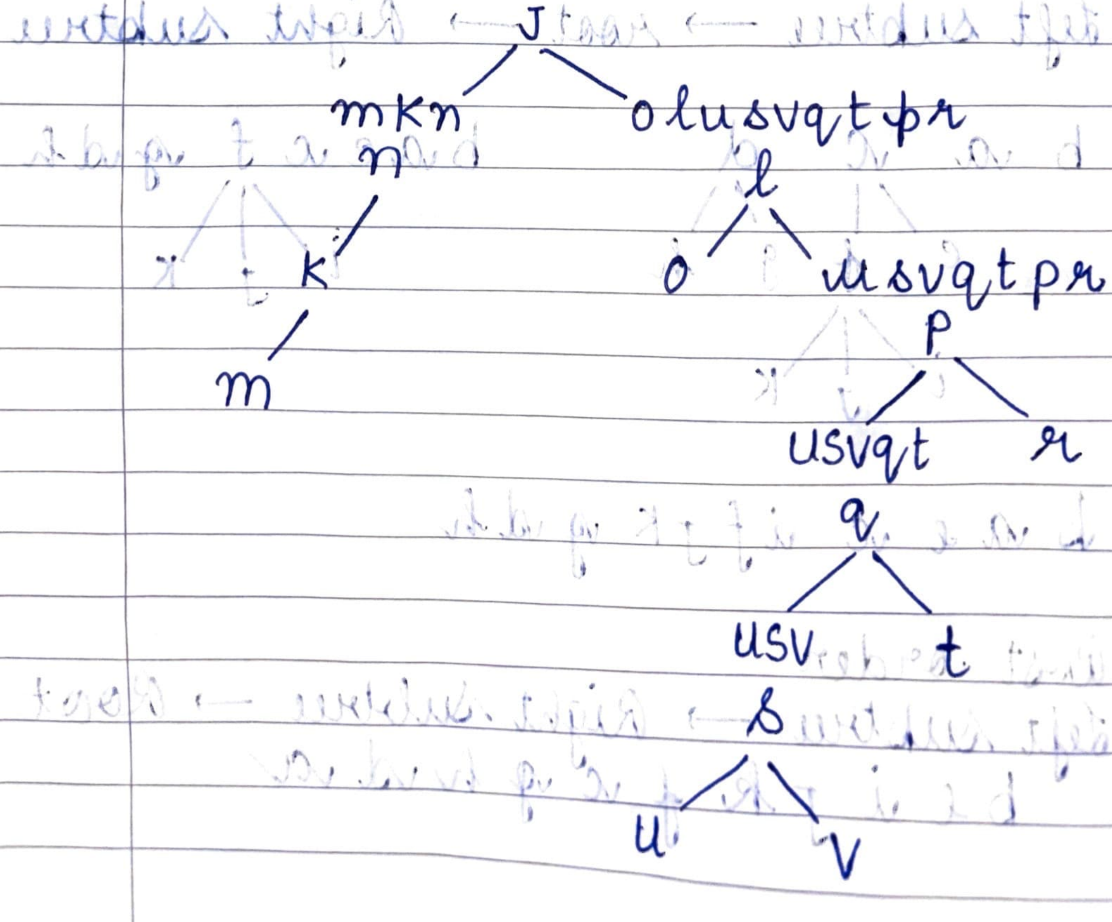
Q ~~Pre~~ In order : $\underbrace{\text{mKn}}_{\text{left}}$ J $\underbrace{\text{o l u s v q t p r}}_{\text{right}}$

Post order : mKnouvstqrplJ

Find tree

Post order ka last element → root

⇒ root → J



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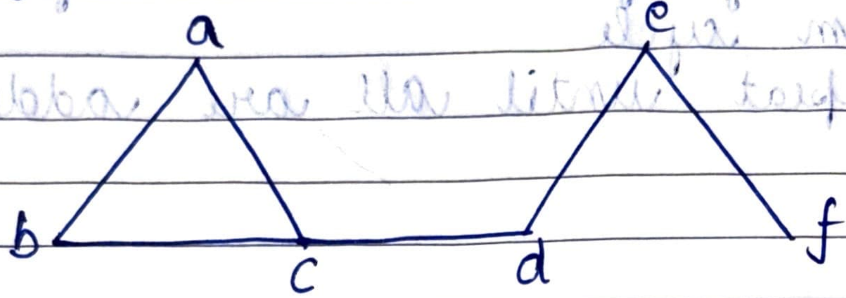
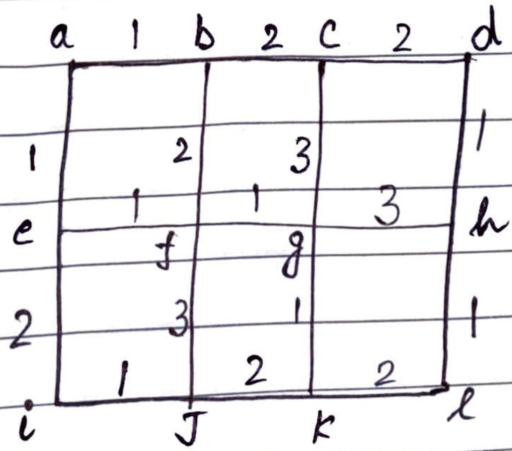
→ Spanning trees

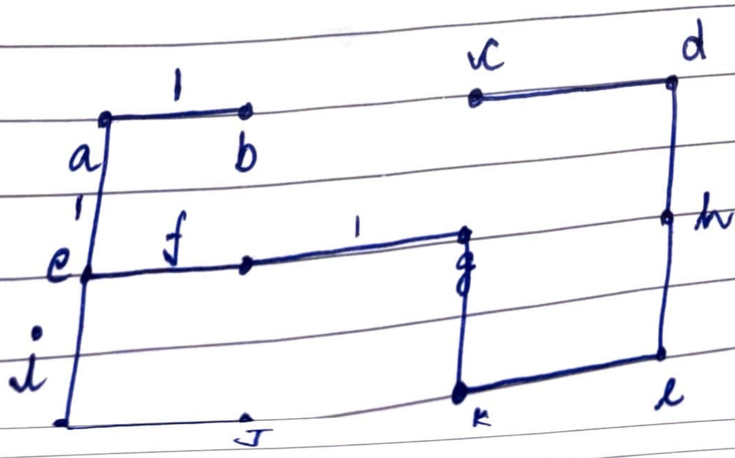
Spanning subgraph of a graph with no cycles



→ Prim's algorithm

- Select an edge with min weight & add to T
- Select edges with min weight & incident to one selected in step-1 & doesn't form cycle
- Repeat until all vertices are added to T





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→ Kruskal's Algorithm

- Arrange all edges in ↑ order of weights
- Add edge from list to T if it doesnt form cycle
- Repeat until all are added